SCSI1113: Computational Mathematics

Sem 1 20212022

ASSIGNMENT 1

1. Let **A**, **B**, and **C** be given by

$$A=\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix} \; ; \; B=\begin{bmatrix} 1 & 2 \\ -2 & -6 \end{bmatrix} \; ; \; C=\begin{bmatrix} 2 & -5 \\ 3 & 4 \end{bmatrix}$$

   a) Find (**AB**)**C** b) Find **A**(**B**+**C**)

2. Find the determinant of the following matrices, if it exists.

$$A=\begin{bmatrix} 2 & 0 & 6 \\ -1 & 5 & -4 \\ 3 & -5 & 2 \end{bmatrix} \; ; \; B=\begin{bmatrix} 2 & 1 & -1 \\ 2 & 2 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

3. Find the inverse of the matrix in Question (2) using adjoint matrix.

4. Find the inverse of the coefficient matrix below using linear system approach.

$$\begin{aligned} -x+y-5z-2w &= -8 \\ x-2y+3z+w &= 5 \\ x-y+5z+3w &= 6 \\ 24y+7z+3w &= 14 \end{aligned}$$

Linear System in Question 8

5. Find the solution of the linear system in Question (4) using Gauss-Jordan method.

6. Solve the following linear equation using Gaussian Elimination method.

$$\begin{aligned} 2.6x+y-z &= -5.4 \\ 3.2x-0.5y+z &= -2.3 \\ -4.6x+y+2z &= 15.9 \end{aligned}$$

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7. Solve the following linear system using LU factorization using [illegible] approach

$$\begin{aligned} 4x_1+6x_2+2x_3 &= -24 \\ 8x_1+4x_2-4x_3 &= -20 \\ 4x_1-2x_2+6x_3 &= 26 \end{aligned}$$

8. Solve the following linear system using LU factorization using Crout approach

$$\begin{aligned} 4x_1+6x_2+2x_3 &= -24 \\ 8x_1+4x_2-4x_3 &= -20 \\ 4x_1-2x_2+6x_3 &= 26 \end{aligned}$$

Before Meeting 2

** Deadline of submission: 7th [illegible] **